

BRYAN JIANG

bryanbj@umich.edu • [Portfolio](#) • [GitHub](#) • [linkedin.com/in/bryanbj15](https://www.linkedin.com/in/bryanbj15)

EDUCATION

University of Michigan, Ann Arbor, MI

Dec 2026

B.S.E. in Computer Science | Minors in Mathematics and Music

GPA: 4.0/4.0

Coursework: Data Structures and Algorithms in C++, Computer Organization, Web Systems, Signal Processing, Computer Networks, Computer Vision, Machine Learning, Database Management, Natural Language Processing

WORK EXPERIENCE

Microsoft, Redmond, WA

May 2026 – August 2026

Applied AI Software Engineer Intern

- Designed and shipped a tab group based context management system for Copilot sessions in Edge, built in C++ on Chromium, enabling parallel AI workflows across browsing contexts for millions of users.
- Orchestrated an multi agent GitHub Copilot pipeline automating bug PR process end-to-end including bug detection and issue filing, PR generation and iteration, style enforced review, and weekly documentation refresh.
- Utilized background loading and copy-on-write caching to reduce edge copilot session load time by 80%.

Here Technology, Ann Arbor, MI

Oct 2025 – Present

Software Engineer

- Developed a data processing pipeline integrating satellite images and geospatial data to automatically detect and route into multiple entry points for large public venues (e.g., airports, stadiums)
- Implemented computer vision and navigation algorithms with Python, Scala, and Java using OpenCV and GeoPandas to identify pedestrian and vehicular routes, improving last-mile navigation accuracy.
- Deployed cloud-based workflows on AWS using S3, Lambda, and EC2 for large-scale data processing and visualization, supporting crowd-aware and accessibility-focused navigation.

Clark Associates, Ann Arbor, MI

May 2025 – Dec 2025

Software Engineer Intern

- Designed and implemented account history management features for Clark's multi-tenant e-commerce application, (Marketplace) using .NET C#, React, and Typescript, enhancing overall user and admin experience.
- Developed MSSQL Server stored procedures, reducing query latency for product filtering/sorting by ~80%.
- Streamlined CI/CD pipelines in Azure DevOps within an Agile development environment.
- Integrated Sentry for real-time error tracking and deployed microservices in Kubernetes using Docker.

PROJECTS

Self-Iterating Strategy Gameplay AI Agent Framework

March 2026 - Present

- Built an AI gameplay agent using Python and the OpenAI API, combining multi-stage LLM reasoning with custom tools and skills with parameterized goal-guided game-tree search to select tactical actions for strategy games.
- Created an analysis pipeline that identifies strategic weaknesses by comparing predicted outcomes and Agent regret, tunes the agent's evaluation model, and stores insights in short and long-term memory compaction system.
- Built an AI evaluation framework with reproducible game-state scenarios, simulated gameplay, and end-to-end decision tests to benchmark alternative agent architectures.

AI-driven Trading Card Game Engine

Sep 2024 - Present

Founder & Software Developer

- Built a web-based AI-driven game engine using Node.js, React, and python, capable of generating TCG rulesets, abilities, and sprite assets from natural language input and automating large-scale simulation testing.
- Designed a custom python SDK and AI agent compiler, generating accurate and complex game system code based on user given text instructions and node-based visual component graphs with inputs, events, and functions.
- Optimized server infrastructure with Redux, Postgres, and asynchronous multi-agent pipelines, reducing API latency by 300% and significantly enhancing real-time responsiveness and scalability.

SKILLS

Programming: C++, C#, Python, Java, .NET, HTML, JavaScript, SQL, React, Git, Julia, OpenGL, MATLAB, Lua

Technologies: Ubuntu Linux, VS Code, Unreal Engine, Unity, Jira, Azure Dev Ops, OpenAI, Logic Pro, Godot, Figma

Interests: Music Production, Game Development, Computer Graphics, Volleyball, Snowboarding, Jazz